Math 20

From UPD IM Module, 2018 — Module 1

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0 Notes

0.1 Quadratics

Note 0.1.1. Definition. Let $a, b, c \in \mathbb{R}$ and $a \neq 0$. A quadratic equation in standard form is:

$$ax^2 + bx + c = 0$$

These equations have either one or two solutions. Note that the square of a number cannot be negative.

Note 0.1.2. Discriminant. Given a quadratic in standard form, its discriminant is:

$$b^2 - 4ac$$

1. If this is greater than zero, the quadratic has two real solutions.
2. If this is equal to zero, the quadratic has one real solution.
3. If this is less than zero, the quadratic has two complex solutions which are conjugates of each other.

Note 0.1.3. Quadratic Formula. Given a quadratic in standard form, the formula to solve for its solutions is:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note 0.1.4. Factoring. Given a quadratic in standard form, you may factor it in order to solve for its solutions. For example:

$$\begin{aligned} 4x^2 + 4x + 1 &= 0 \\ (2x+1)^2 &= 0 \\ &\boxed{x = -\frac{1}{2}} \end{aligned}$$

Note 0.1.5. Equations in Quadratic Form. Given the equation:

$$2\left(x + \frac{1}{x}\right)^2 + \left(x + \frac{1}{x}\right) - 10 = 0$$

While not a quadratic equation, we may rewrite it as such:

$$2t^2 + t - 10 = 0, \text{ where } t = x + \frac{1}{x}$$

Solve for the values of t as usual, then solve for x using t. For example, using the equation above:

$$2\left(x + \frac{1}{x}\right)^2 + \left(x + \frac{1}{x}\right) - 10 = 0$$

$t = x + \dfrac{1}{x}$

$$\begin{aligned} 2t^2 + t - 10 &= 0 \\ 2t^2 + 5t - 4t - 10 &= 0 \\ (t-2)(2t+5) &= 0 \\ t &\in \left\{-\frac{5}{2}, 2\right\} \end{aligned}$$

Solve for x, $t = -\dfrac{5}{2}$.

$$x + \frac{1}{x} = -\frac{5}{2}$$

$x \neq 0$

$$\begin{aligned} 2x^2 + 5x + 2 &= 0 \\ 2x^2 + 4x + x + 2 &= 0 \\ (2x+1)(x+2) &= 0 \\ x &\in \left\{-2, -\frac{1}{2}\right\} \end{aligned}$$

Solve for x, $t = 2$.

$$x + \frac{1}{x} = 2$$

$x \neq 0$

$$\begin{aligned} x^2 - 2x + 1 &= 0 \\ (x-1)^2 &= 0 \\ x &= 1 \\ &\boxed{x \in \left\{-2, -\frac{1}{2}, 1\right\}} \end{aligned}$$

0.2 Radicals

Note 0.2.1. Negative Numbers. Note that $\sqrt[n]{x}$ cannot be negative for even values of n.

Note 0.2.2. Extraneous Solutions. It is required to check each solution when exponentiating both sides of an equation, as these may be extraneous solutions, which are not solutions to the original equation.

0.3 Absolute Values

Note 0.3.1. Definition.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Note that the absolute value of a number cannot be negative.

Note 0.3.2. Of Products.

$$|xy| = |x||y|$$

Note 0.3.3. Of Quotients.

$$\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$$

0.4 Inequalities

Note 0.4.1. Quadratics. In order to solve quadratic inequalities, first rewrite it in standard form, factor, identify critical numbers and undefined points, then create a table of signs.

Note 0.4.2. Absolute Values. In order to solve inequalities with absolute values, take note of the following:

1. $|x| < a$ if and only if $-a < x < a$.
2. $|x| > a$ if and only if $x < -a$ or $x > a$.

1

Solve the following equations.

1.1

$$x + 9 = 5 - 3x$$

$$4x = -4$$

$$\boxed{x = -1}$$

1.2

$$\frac{2x+3}{4} - \frac{x+1}{2} = -\frac{1}{3}$$

$$\begin{aligned} 6x + 9 - 6x - 6 &= -4 \\ 3 &= -4 \end{aligned}$$

$$\boxed{x \in \emptyset}$$

1.3

$$3x^2 - 2x + 1 = 0$$

Note 0.1.3

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(3)(1)}}{2(3)}$$
$$x = \frac{2 \pm \sqrt{4 - 12}}{6}$$
$$x = \frac{2 \pm \sqrt{-8}}{6}$$
$$x = \frac{2 \pm 2i\sqrt{2}}{6}$$
$$\boxed{x = \frac{1 \pm i\sqrt{2}}{3}}$$

1.4

$$4x^2 + 2x = 2$$

$$\begin{aligned} 2x^2 + x - 1 &= 0 \\ 2x^2 + 2x - x - 1 &= 0 \\ (2x-1)(x+1) &= 0 \end{aligned}$$

$$\boxed{x \in \left\{-1, \frac{1}{2}\right\}}$$

1.5

$$16x^2 + 9 = 24x$$

$$\begin{aligned} 16x^2 - 24x + 9 &= 0 \\ (4x-3)^2 &= 0 \end{aligned}$$

$$\boxed{x = \frac{3}{4}}$$

1.6

$$\frac{x}{x-1} + \frac{x-5}{x^2+2x-3} = \frac{1}{x+3}$$

$$\frac{x}{x-1} + \frac{x-5}{(x-1)(x+3)} = \frac{1}{x+3}$$

$x \notin \{-3, 1\}$

$$\begin{aligned} x^2 + 3x + x - 5 &= x - 1 \\ x^2 + 3x - 4 &= 0 \\ (x+4)(x-1) &= 0 \\ x &\in \{-4, 1\} \end{aligned}$$

$$\boxed{x = -4}$$

1.7

$$\frac{1}{4-x} + \frac{3}{6+x} = 0$$

$x \notin \{-6, 4\}$

$$\begin{aligned} 6 + x + 12 - 3x &= 0 \\ 2x &= 18 \end{aligned}$$

$$\boxed{x = 9}$$

1.8

$$\frac{3}{x^2-9} - \frac{7}{x-3} = -\frac{4}{x+3}$$

$$\frac{3}{(x-3)(x+3)} - \frac{7}{x-3} = -\frac{4}{x+3}$$

$x \notin \{-3, 3\}$

$$\begin{aligned} 3 - 7x - 21 &= -4x + 12 \\ 3x &= -30 \end{aligned}$$

$$\boxed{x = -10}$$

1.9

$$\frac{3}{x^2 - x - 6} = \frac{4}{2x^2 + x - 6}$$

$$\frac{3}{(x-3)(x+2)} = \frac{4}{2x^2 + 4x - 3x - 6}$$
$$\frac{3}{(x-3)(x+2)} = \frac{4}{(2x-3)(x+2)}$$

$x \notin \left\{-2, \frac{3}{2}, 3\right\}$

$$6x - 9 = 4x - 12$$
$$2x = -3$$

$$\boxed{x = -\frac{3}{2}}$$

1.10

$$\frac{2}{y+1} - \frac{3}{1-y} = \frac{5}{y}$$

$y \notin \{-1, 0, 1\}$

$$\begin{aligned} 2y(1-y) - 3y(y+1) &= 5(y+1)(1-y) \\ 2y - 2y^2 - 3y^2 - 3y &= 5(y - y^2 + 1 - y) \\ -5y^2 - y &= -5y^2 + 5 \end{aligned}$$

$$\boxed{y = -5}$$

1.11

$$\frac{x+17}{x^2-6x+8}+\frac{x-2}{x-4}=\frac{x-4}{x-2}$$

$$\frac{x+17}{(x-4)(x-2)}+\frac{x-2}{x-4}=\frac{x-4}{x-2}$$

$x \notin \{2,4\}$

$$\begin{aligned} x+17+(x-2)^2 &= (x-4)^2 \\ x+17+x^2-4x+4 &= x^2-8x+16 \\ 5x &= -5 \end{aligned}$$

$$\boxed{x=-1}$$

1.12

$$\frac{w}{3w^2 - 8w + 4} = \frac{w+2}{3w^2 + w - 2}$$

$$\frac{w}{3w^2 - 6w - 2w + 4} = \frac{w+2}{3w^2 + 3w - 2w - 2}$$
$$\frac{w}{(3w-2)(w-2)} = \frac{w+2}{(3w-2)(w+1)}$$

$w \notin \left\{-1, \frac{2}{3}, 2\right\}$

$$w^2 + w = w^2 - 4$$

$$\boxed{w = -4}$$

1.13

$$\frac{3}{x+4} = \frac{2}{3x-2}$$

$x \notin \left\{-4, \frac{2}{3}\right\}$

$$9x - 6 = 2x + 8$$
$$7x = 14$$

$$\boxed{x = 2}$$

1.14

$$\frac{x+4}{x+2} = \frac{2x-3}{x-5}$$

$x \notin \{-2, 5\}$

$$\begin{aligned}(x+4)(x-5) &= (2x-3)(x+2)\\ x^2 - 5x + 4x - 20 &= 2x^2 + 4x - 3x - 6\\ x^2 + 2x + 14 &= 0\end{aligned}$$

Note 0.1.3

$$\begin{aligned}x &= \frac{-2 \pm \sqrt{2^2 - 4(1)(14)}}{2(1)}\\ x &= \frac{-2 \pm \sqrt{4 - 56}}{2}\\ x &= \frac{-2 \pm \sqrt{-52}}{2}\\ x &= \frac{-2 \pm 2i\sqrt{13}}{2}\end{aligned}$$

$$\boxed{x = -1 \pm i\sqrt{13}}$$

1.15

$$\frac{4}{z-2} = \frac{z}{6}$$

$z \neq 2$

$$\begin{aligned} 24 &= z^2 - 2z \\ z^2 - 2z - 24 &= 0 \\ (z-6)(z+4) &= 0 \end{aligned}$$

$$\boxed{z \in \{-4, 6\}}$$

1.16

$$\frac{1}{2x+5} = \frac{4}{2x-1}$$

$x \notin \left\{-\frac{5}{2}, \frac{1}{2}\right\}$

$$\begin{aligned} 2x - 1 &= 8x + 20 \\ 6x &= -21 \\ x &= -\frac{21}{6} \end{aligned}$$

$$\boxed{x = -\frac{7}{2}}$$

1.17

$$\frac{2}{3x} + \frac{4}{5x} = \frac{11}{15}$$

$x \neq 0$

$$10 + 12 = 11x$$
$$11x = 22$$

$$\boxed{x = 2}$$

1.18

$$\frac{5}{x+3} = \frac{x+8}{x+3} - 3$$

$x \neq -3$

$$\begin{aligned} 5 &= x + 8 - 3x - 9 \\ 2x &= -6 \\ x &= -3 \end{aligned}$$

$$\boxed{x \in \emptyset}$$

1.19

$$\frac{3x}{x+2} = \frac{x+4}{x}$$

$x \notin \{-2, 0\}$

$$\begin{aligned} 3x^2 &= (x+4)(x+2) \\ 3x^2 &= x^2 + 2x + 4x + 8 \\ 2x^2 - 6x - 8 &= 0 \\ x^2 - 3x - 4 &= 0 \\ (x-4)(x+1) &= 0 \end{aligned}$$

$$\boxed{x \in \{-1, 4\}}$$

1.20

$$\frac{x+4}{x+2} - \frac{2x-3}{x-5} = \frac{3x-8}{x^2-3x-10}$$

$$\frac{x+4}{x+2} - \frac{2x-3}{x-5} = \frac{3x-8}{(x+2)(x-5)}$$

$x \notin \{-2, 5\}$

$$\begin{aligned}
(x+4)(x-5) - (2x-3)(x+2) &= 3x-8 \\
x^2 - 5x + 4x - 20 - 2x^2 - 4x + 3x + 6 &= 3x-8 \\
x^2 + 5x + 6 &= 0 \\
(x+3)(x+2) &= 0 \\
x &\in \{-3, -2\}
\end{aligned}$$

$$\boxed{x = -3}$$

1.21

$$1 + \frac{5x}{6} = \frac{2x^2}{3}$$

$$\begin{aligned} 6 + 5x &= 4x^2 \\ 4x^2 - 5x - 6 &= 0 \\ 4x^2 - 8x + 3x - 6 &= 0 \\ (4x+3)(x-2) &= 0 \end{aligned}$$

$$\boxed{x \in \left\{-\frac{3}{4}, 2\right\}}$$

1.22

$$\frac{x}{2x+3} = x + 4$$

$x \neq -\frac{3}{2}$

$$\begin{aligned} x &= 2x^2 + 3x + 8x + 12 \\ 2x^2 + 10x + 12 &= 0 \\ x^2 + 5x + 6 &= 0 \\ (x+3)(x+2) &= 0 \end{aligned}$$

$$\boxed{x \in \{-3, -2\}}$$

1.23

$$\sqrt{x} + 5 = 3$$

$$\sqrt{x} = -2$$

Note 0.2.1

$$\boxed{x \in \emptyset}$$

1.24

$$\sqrt{3x+7} - x = 1$$

$$\begin{aligned}
\sqrt{3x+7} &= x+1 \\
3x+7 &= x^2+2x+1 \\
x^2-x-6 &= 0 \\
(x-3)(x+2) &= 0 \\
x &\in \{-2, 3\}
\end{aligned}$$

Note 0.2.2, $x = -2$.

$$\begin{aligned}
\sqrt{3(-2)+7} + 2 &= 1 \\
3 &\neq 1
\end{aligned}$$

Note 0.2.2, $x = 3$.

$$\begin{aligned}
\sqrt{3(3)+7} - 3 &= 1 \\
1 &= 1
\end{aligned}$$

$$\boxed{x = 3}$$

1.25

$$\sqrt{\sqrt{5x-1}-x}=1$$

$$\begin{aligned}
\sqrt{5x-1}-x&=1\\
\sqrt{5x-1}&=x+1\\
5x-1&=x^2+2x+1\\
x^2-3x+2&=0\\
(x-2)(x-1)&=0\\
x&\in\{1,2\}
\end{aligned}$$

Note 0.2.2, $x=1$.

$$\begin{aligned}
\sqrt{\sqrt{5(1)-1}-1}&=1\\
1&=1
\end{aligned}$$

Note 0.2.2, $x=2$.

$$\begin{aligned}
\sqrt{\sqrt{5(2)-1}-2}&=1\\
1&=1
\end{aligned}$$

$$\boxed{x\in\{1,2\}}$$

1.26

$$\sqrt{5w+1} - \sqrt{3w} - 1 = 0$$

$$\begin{aligned}
\sqrt{5w+1} &= \sqrt{3w} + 1 \\
5w + 1 &= 3w + 2\sqrt{3w} + 1 \\
\sqrt{3w} &= w \\
w^2 - 3w &= 0 \\
w(w-3) &= 0 \\
w &\in \{0, 3\}
\end{aligned}$$

Note 0.2.2, $w = 0$.

$$\begin{aligned}
\sqrt{5(0)+1} - \sqrt{3(0)} - 1 &= 0 \\
0 &= 0
\end{aligned}$$

Note 0.2.2, $w = 3$.

$$\begin{aligned}
\sqrt{5(3)+1} - \sqrt{3(3)} - 1 &= 0 \\
0 &= 0
\end{aligned}$$

$$\boxed{w \in \{0, 3\}}$$

1.27

$$\sqrt{x-2} = 2$$

$$\begin{aligned} x - 2 &= 4 \\ x &= 6 \end{aligned}$$

Note 0.2.2, $x = 6$.

$$\begin{aligned} \sqrt{6-2} &= 2 \\ 2 &= 2 \end{aligned}$$

$$\boxed{x = 6}$$

1.28

$$\sqrt{2x+5}+x=5$$

$$\begin{aligned}\sqrt{2x+5}&=5-x\\2x+5&=x^2-10x+25\\x^2-12x+20&=0\\(x-10)(x-2)&=0\\x&\in\{2,10\}\end{aligned}$$

Note 0.2.2, $x=2$.

$$\begin{aligned}\sqrt{2(2)+5}+2&=5\\5&=5\end{aligned}$$

Note 0.2.2, $x=10$.

$$\begin{aligned}\sqrt{2(10)+5}+10&=5\\15&\neq 5\end{aligned}$$

$$\boxed{x=2}$$

1.29

$$\sqrt{x^2 - 8x} = 3$$

$$\begin{aligned} x^2 - 8x - 9 &= 0 \\ (x-9)(x+1) &= 0 \\ x &\in \{-1, 9\} \end{aligned}$$

Note 0.2.2, $x = -1$.

$$\begin{aligned} \sqrt{(-1)^2 - 8(-1)} &= 3 \\ 3 &= 3 \end{aligned}$$

Note 0.2.2, $x = 9$.

$$\begin{aligned} \sqrt{9^2 - 8(9)} &= 3 \\ 3 &= 3 \end{aligned}$$

$$\boxed{x \in \{-1, 9\}}$$

1.30

$$\sqrt{x+\sqrt{x-2}}=2$$

$$\begin{aligned}
x+\sqrt{x-2}&=4\\
\sqrt{x-2}&=4-x\\
x-2&=x^2-8x+16\\
x^2-9x+18&=0\\
(x-6)(x-3)&=0\\
x&\in\{3,6\}
\end{aligned}$$

Note 0.2.2, $x=3$.

$$\begin{aligned}
\sqrt{3+\sqrt{3-2}}&=2\\
2&=2
\end{aligned}$$

Note 0.2.2, $x=6$.

$$\begin{aligned}
\sqrt{6+\sqrt{6-2}}&=2\\
2\sqrt{2}&\neq 2
\end{aligned}$$

$$\boxed{x=3}$$

1.31

$$\sqrt{x-1} - \sqrt{x-2} - 1 = 0$$

$$\begin{aligned}
\sqrt{x-1} - 1 &= \sqrt{x-2} \\
x - 1 - 2\sqrt{x-1} + 1 &= x - 2 \\
2\sqrt{x-1} &= 2 \\
4x - 4 &= 4 \\
4x &= 8 \\
x &= 2
\end{aligned}$$

Note 0.2.2, $x = 2$.

$$\begin{aligned}
\sqrt{2-1} - \sqrt{2-2} - 1 &= 0 \\
0 &= 0
\end{aligned}$$

$$\boxed{x = 2}$$

1.32

$$2\sqrt{x-1}-\sqrt{x-2}-2=0$$

$$\begin{aligned}
2\sqrt{x-1}-2&=\sqrt{x-2}\\
4x-4-8\sqrt{x-1}+4&=x-2\\
3x+2&=8\sqrt{x-1}\\
9x^2+12x+4&=64x-64\\
9x^2-52x+68&=0
\end{aligned}$$

Note 0.1.3

$$\begin{aligned}
x&=\frac{52\pm\sqrt{(-52)^2-4(9)(68)}}{2(9)}\\
x&=\frac{52\pm 16}{18}\\
x&=\frac{26\pm 8}{9}\\
x&\in\left\{\frac{18}{9},\frac{34}{9}\right\}\\
x&\in\left\{2,\frac{34}{9}\right\}
\end{aligned}$$

Note 0.2.2, $x=2$.

$$\begin{aligned}
2\sqrt{2-1}-\sqrt{2-2}-2&=0\\
0&=0
\end{aligned}$$

Note 0.2.2, $x=\dfrac{34}{9}$.

$$\begin{aligned}
2\sqrt{\frac{34}{9}-1}-\sqrt{\frac{34}{9}-2}-2&=0\\
2\sqrt{\frac{25}{9}}-\sqrt{\frac{16}{9}}-2&=0\\
\frac{10}{3}-\frac{4}{3}-\frac{6}{3}&=0\\
0&=0
\end{aligned}$$

$$\boxed{x\in\left\{2,\frac{34}{9}\right\}}$$

1.33

$$\sqrt{3-3x}-\sqrt{3x+2}=3$$

$$\begin{aligned}
\sqrt{3-3x} &= \sqrt{3x+2}+3 \\
3-3x &= 3x+2+6\sqrt{3x+2}+9 \\
-6x-8 &= 6\sqrt{3x+2} \\
-3x-4 &= 3\sqrt{3x+2} \\
9x^2+24x+16 &= 27x+18 \\
9x^2-3x-2 &= 0 \\
9x^2-6x+3x-2 &= 0 \\
(3x+1)(3x-2) &= 0 \\
x &\in \left\{-\frac{1}{3}, \frac{2}{3}\right\}
\end{aligned}$$

Note 0.2.2, $x=-\frac{1}{3}$.

$$\begin{aligned}
\sqrt{3-3\left(-\frac{1}{3}\right)}-\sqrt{3\left(-\frac{1}{3}\right)+2} &= 3 \\
\sqrt{3+1}-\sqrt{-1+2} &= 3 \\
1 &\neq 3
\end{aligned}$$

Note 0.2.2, $x=\frac{2}{3}$.

$$\begin{aligned}
\sqrt{3-3\left(\frac{2}{3}\right)}-\sqrt{3\left(\frac{2}{3}\right)+2} &= 3 \\
\sqrt{3-2}-\sqrt{2+2} &= 3 \\
-1 &\neq 3
\end{aligned}$$

$$\boxed{x \in \emptyset}$$

1.34

$$2 + 2\sqrt[3]{x-1} = x$$

$$\begin{aligned}
2\sqrt[3]{x-1} &= x - 2 \\
8x - 8 &= x^3 - 6x^2 + 12x - 8 \\
x^3 - 6x^2 + 4x &= 0 \\
x(x^2 - 6x + 4) &= 0
\end{aligned}$$

$x = 0$, solve for $x^2 - 6x + 4 = 0$, note 0.1.3.

$$\begin{aligned}
x &= \frac{6 \pm \sqrt{(-6)^2 - 4(1)(4)}}{2(1)} \\
x &= \frac{6 \pm \sqrt{36 - 16}}{2} \\
x &= \frac{6 \pm \sqrt{20}}{2} \\
x &= \frac{6 \pm 2\sqrt{5}}{2} \\
x &= 3 \pm \sqrt{5} \\
x &\in \{0, 3 \pm \sqrt{5}\}
\end{aligned}$$

Note 0.2.2, $x = 0$.

$$\begin{aligned}
2 + 2\sqrt[3]{0-1} &= 0 \\
0 &= 0
\end{aligned}$$

Note 0.2.2, $x = 3 + \sqrt{5}$.

$$\begin{aligned}
2 + 2\sqrt[3]{3 + \sqrt{5} - 1} &= 3 + \sqrt{5} \\
2\sqrt[3]{2 + \sqrt{5}} &= 1 + \sqrt{5} \\
16 + 8\sqrt{5} &= 1 + 3\sqrt{5} + 15 + 5\sqrt{5} \\
16 + 8\sqrt{5} &= 16 + 8\sqrt{5}
\end{aligned}$$

Note 0.2.2, $x = 3 - \sqrt{5}$.

$$\begin{aligned}
2 + 2\sqrt[3]{3 - \sqrt{5} - 1} &= 3 - \sqrt{5} \\
2\sqrt[3]{2 - \sqrt{5}} &= 1 - \sqrt{5} \\
16 - 8\sqrt{5} &= 1 - 3\sqrt{5} + 15 - 5\sqrt{5} \\
16 - 8\sqrt{5} &= 16 - 8\sqrt{5}
\end{aligned}$$

$$\boxed{x \in \{0, 3 \pm \sqrt{5}\}}$$

1.35

$$\sqrt[4]{3x^2 - 3x + 1} = \sqrt{x - 2}$$

$$\begin{aligned} 3x^2 - 3x + 1 &= x^2 - 4x + 4 \\ 2x^2 + x - 3 &= 0 \\ 2x^2 + 3x - 2x - 3 &= 0 \\ (x-1)(2x+3) &= 0 \\ x &\in \left\{-\frac{3}{2}, 1\right\} \end{aligned}$$

Note 0.2.2, $x = -\frac{3}{2}$.

$$\begin{aligned} \sqrt[4]{3\left(-\frac{3}{2}\right)^2 - 3\left(-\frac{3}{2}\right) + 1} &= \sqrt{-\frac{3}{2} - 2} \\ \sqrt[4]{3\left(\frac{9}{4}\right) + \frac{18}{4} + \frac{4}{4}} &= i\sqrt{\frac{7}{2}} \\ \sqrt[4]{\frac{27}{4} + \frac{18}{4} + \frac{4}{4}} &= i\sqrt{\frac{7}{2}} \\ \sqrt[4]{\frac{49}{4}} &= i\sqrt{\frac{7}{2}} \\ \sqrt{\frac{7}{2}} &\neq i\sqrt{\frac{7}{2}} \end{aligned}$$

Note 0.2.2, $x = 1$.

$$\begin{aligned} \sqrt[4]{3(1)^2 - 3(1) + 1} &= \sqrt{1 - 2} \\ \sqrt[4]{3 - 3 + 1} &= i \\ 1 &\neq i \end{aligned}$$

$$\boxed{x \in \emptyset}$$

1.36

$$\sqrt{2x+3} - \sqrt{x-2} = \sqrt{x+1}$$

$$\begin{aligned}
2x + 3 - 2\sqrt{2x+3}\sqrt{x-2} + x - 2 &= x + 1 \\
2x &= 2\sqrt{2x+3}\sqrt{x-2} \\
4x^2 &= 4(2x^2 - 4x + 3x - 6) \\
4x^2 &= 8x^2 - 16x + 12x - 24 \\
4x^2 - 4x - 24 &= 0 \\
x^2 - x - 6 &= 0 \\
(x-3)(x+2) &= 0 \\
x &\in \{-2, 3\}
\end{aligned}$$

Note 0.2.2, $x = -2$.

$$\begin{aligned}
\sqrt{2(-2)+3} - \sqrt{-2-2} &= \sqrt{-2+1} \\
i - 2i &= i \\
-i &\neq i
\end{aligned}$$

Note 0.2.2, $x = 3$.

$$\begin{aligned}
\sqrt{2(3)+3} - \sqrt{3-2} &= \sqrt{3+1} \\
2 &= 2
\end{aligned}$$

$$\boxed{x = 3}$$

1.37

$$1 = x + \sqrt{2x - 3}$$

$$\begin{aligned} 1 - x &= \sqrt{2x - 3} \\ x^2 - 2x + 1 &= 2x - 3 \\ x^2 - 4x + 4 &= 0 \\ (x - 2)^2 &= 0 \\ x &= 2 \end{aligned}$$

Note 0.2.2, $x = 2$.

$$\begin{aligned} 1 &= 2 + \sqrt{2(2) - 3} \\ 1 &\neq 3 \end{aligned}$$

$$\boxed{x \in \emptyset}$$

1.38

$$8\left(x+\frac{1}{2x}\right)^2+30\left(x+\frac{1}{2x}\right)+27=0$$

Note 0.1.5, $t=x+\dfrac{1}{2x}$.

$$8t^2+30t+27=0$$

Note 0.1.3

$$\begin{aligned}
t&=\frac{-30\pm\sqrt{30^2-4(8)(27)}}{2(8)}\\
t&=\frac{-30\pm\sqrt{900-864}}{16}\\
t&=\frac{-30\pm\sqrt{36}}{16}\\
t&=\frac{-30\pm 6}{16}\\
t&=\frac{-15\pm 3}{8}\\
t&\in\left\{-\frac{18}{8},-\frac{12}{8}\right\}
\end{aligned}$$

Note 0.1.5, $t=-\dfrac{18}{8}$.

$$x+\frac{1}{2x}=-\frac{18}{8}$$

$x\neq 0$

$$\begin{aligned}
8x^2+4&=-18x\\
4x^2+9x+2&=0\\
4x^2+8x+x+2&=0\\
(4x+1)(x+2)&=0\\
x&\in\left\{-2,-\frac{1}{4}\right\}
\end{aligned}$$

Note 0.1.5, $t=-\dfrac{12}{8}$.

$$x+\frac{1}{2x}=-\frac{12}{8}$$

$x\neq 0$

$$\begin{aligned}
8x^2+4&=-12x\\
2x^2+3x+1&=0\\
2x^2+2x+x+1&=0\\
(2x+1)(x+1)&=0\\
x&\in\left\{-1,-\frac{1}{2}\right\}
\end{aligned}$$

$$\boxed{x\in\left\{-2,-1,-\frac{1}{2},-\frac{1}{4}\right\}}$$

1.39

$$\sqrt[4]{x} + 2\sqrt{x} = 3$$

Note 0.1.5, $t = \sqrt[4]{x}$.

$$\begin{aligned} 2t^2 + t - 3 &= 0 \\ 2t^2 + 3t - 2t - 3 &= 0 \\ (t-1)(2t+1) &= 0 \\ t &\in \left\{-\frac{1}{2}, 1\right\} \end{aligned}$$

Note 0.1.5, $t = -\frac{1}{2}$.

$$\sqrt[4]{x} = -\frac{1}{2}$$

Note 0.2.1

$$x \in \emptyset$$

Note 0.1.5, $t = 1$.

$$\begin{aligned} \sqrt[4]{x} &= 1 \\ x &= 1 \end{aligned}$$

Note 0.2.2, $x = 1$.

$$\begin{aligned} \sqrt[4]{1} &= 1 \\ 1 &= 1 \end{aligned}$$

$$\boxed{x = 1}$$

1.40

$$\sqrt{x} - 5\sqrt[4]{x} + 4 = 0$$

Note 0.1.5, $t = \sqrt[4]{x}$.

$$\begin{aligned} t^2 - 5t + 4 &= 0 \\ (t-4)(t-1) &= 0 \\ t &\in \{1, 4\} \end{aligned}$$

Note 0.1.5, $t = 1$.

$$\begin{aligned} \sqrt[4]{x} &= 1 \\ x &= 1 \end{aligned}$$

Note 0.2.2, $x = 1$.

$$\begin{aligned} \sqrt[4]{1} &= 1 \\ 1 &= 1 \end{aligned}$$

Note 0.1.5, $t = 4$.

$$\begin{aligned} \sqrt[4]{x} &= 4 \\ x &= 256 \end{aligned}$$

Note 0.2.2, $x = 256$.

$$\begin{aligned} \sqrt[4]{256} &= 4 \\ 4 &= 4 \end{aligned}$$

$$\boxed{x \in \{1, 256\}}$$

1.41

$$\frac{15}{t^2} - \frac{14}{t} - 8 = 0$$

Note 0.1.5, $u = \frac{1}{t}$.

$$15u^2 - 14u - 8 = 0$$

Note 0.1.3

$$\begin{aligned}
u &= \frac{14 \pm \sqrt{(-14)^2 - 4(15)(-8)}}{2(15)} \\
u &= \frac{14 \pm \sqrt{196 + 480}}{30} \\
u &= \frac{14 \pm \sqrt{676}}{30} \\
u &= \frac{7 \pm 13}{15} \\
u &\in \left\{-\frac{6}{15}, \frac{20}{15}\right\} \\
u &\in \left\{-\frac{2}{5}, \frac{4}{3}\right\}
\end{aligned}$$

Note 0.1.5, $u = -\frac{2}{5}$.

$$\frac{1}{t} = -\frac{2}{5}$$

$t \neq 0$

$$t = -\frac{5}{2}$$

Note 0.1.5, $u = \frac{4}{3}$.

$$\frac{1}{t} = \frac{4}{3}$$

$t \neq 0$

$$t = \frac{3}{4}$$

$$\boxed{t \in \left\{-\frac{5}{2}, \frac{3}{4}\right\}}$$

1.42

$$t^4 - 4 = 0$$

$$\begin{aligned}(t^2 - 2)(t^2 + 2) &= 0 \\ (t - \sqrt{2})(t + \sqrt{2})(t - i\sqrt{2})(t + i\sqrt{2}) &= 0\end{aligned}$$

$$\boxed{t \in \{\pm\sqrt{2}, \pm i\sqrt{2}\}}$$

1.43

$$x^4 - 2x^2 - 15 = 0$$

$$\begin{aligned}(x^2 - 5)(x^2 + 3) &= 0 \\ (x - \sqrt{5})(x + \sqrt{5})(x - i\sqrt{3})(x + i\sqrt{3}) &= 0\end{aligned}$$

$$\boxed{x \in \{\pm\sqrt{5}, \pm i\sqrt{3}\}}$$

1.44

$$2\left(2x+\frac{3}{2}\right)^2-7\left(2x+\frac{3}{2}\right)+3=0$$

Note 0.1.5, $t=2x+\frac{3}{2}$.

$$\begin{aligned}2t^2-7t+3&=0\\2t-6t-t+3&=0\\(2t-1)(t-3)&=0\\t&\in\left\{\frac{1}{2},3\right\}\end{aligned}$$

Note 0.1.5, $t=\frac{1}{2}$.

$$\begin{aligned}2x+\frac{3}{2}&=\frac{1}{2}\\4x+3&=1\\4x&=-2\\x&=-\frac{1}{2}\end{aligned}$$

Note 0.1.5, $t=3$.

$$\begin{aligned}2x+\frac{3}{2}&=3\\4x+3&=6\\4x&=3\\x&=\frac{3}{4}\end{aligned}$$

$$\boxed{x\in\left\{-\frac{1}{2},\frac{3}{4}\right\}}$$

1.45

$$2x^{\frac{2}{3}} - 5x^{\frac{1}{3}} - 3 = 0$$

$$2\sqrt[3]{x^2} - 5\sqrt[3]{x} - 3 = 0$$

Note 0.1.5, $t = \sqrt[3]{x}$.

$$\begin{aligned} 2t^2 - 5t - 3 &= 0 \\ 2t^2 - 6t + t - 3 &= 0 \\ (2t+1)(t-3) &= 0 \\ t &\in \left\{-\frac{1}{2}, 3\right\} \end{aligned}$$

Note 0.1.5, $t = -\frac{1}{2}$.

$$\begin{aligned} \sqrt[3]{x} &= -\frac{1}{2} \\ x &= -\frac{1}{8} \end{aligned}$$

Note 0.2.2, $x = -\frac{1}{8}$.

$$\begin{aligned} \sqrt[3]{-\frac{1}{8}} &= -\frac{1}{2} \\ -\frac{1}{2} &= -\frac{1}{2} \end{aligned}$$

Note 0.1.5, $t = 3$.

$$\begin{aligned} \sqrt[3]{x} &= 3 \\ x &= 27 \end{aligned}$$

Note 0.2.2, $x = 27$.

$$\begin{aligned} \sqrt[3]{27} &= 3 \\ 3 &= 3 \end{aligned}$$

$$\boxed{x \in \left\{-\frac{1}{8}, 27\right\}}$$

1.46

$$3\left(4x-\frac{1}{4}\right)^2-4\left(4x-\frac{1}{4}\right)-15=0$$

Note 0.1.5, $t=4x-\frac{1}{4}$.

$$\begin{aligned}3t^2-4t-15&=0\\3t^2-9t+5t-15&=0\\(3t+5)(t-3)&=0\\t&\in\left\{-\frac{5}{3},3\right\}\end{aligned}$$

Note 0.1.5, $t=-\frac{5}{3}$.

$$\begin{aligned}4x-\frac{1}{4}&=-\frac{5}{3}\\48x-3&=-20\\48x&=-17\\x&=-\frac{17}{48}\end{aligned}$$

Note 0.1.5, $t=3$.

$$\begin{aligned}4x-\frac{1}{4}&=3\\16x-1&=12\\16x&=13\\x&=\frac{13}{16}\end{aligned}$$

$$\boxed{x\in\left\{-\frac{17}{48},\frac{13}{16}\right\}}$$

1.47

$$6 + \sqrt[3]{x+1} = \sqrt[3]{(x+1)^2}$$

$$\sqrt[3]{(x+1)^2} - \sqrt[3]{x+1} - 6 = 0$$

Note 0.1.5, $t = \sqrt[3]{x+1}$.

$$\begin{aligned} t^2 - t - 6 &= 0 \\ (t-3)(t+2) &= 0 \\ t &\in \{-2, 3\} \end{aligned}$$

Note 0.1.5, $t = -2$.

$$\begin{aligned} \sqrt[3]{x+1} &= -2 \\ x+1 &= -8 \\ x &= -9 \end{aligned}$$

Note 0.2.2, $x = -9$.

$$\begin{aligned} \sqrt[3]{-9+1} &= -2 \\ -2 &= -2 \end{aligned}$$

Note 0.1.5, $t = 3$.

$$\begin{aligned} \sqrt[3]{x+1} &= 3 \\ x+1 &= 27 \\ x &= 26 \end{aligned}$$

Note 0.2.2, $x = 26$.

$$\begin{aligned} \sqrt[3]{26+1} &= 3 \\ 3 &= 3 \end{aligned}$$

$$\boxed{x \in \{-9, 26\}}$$

1.48

$$-7\left(\frac{1}{x}-1\right)=4-2\left(\frac{1}{x}-1\right)^2$$

$$2\left(\frac{1}{x}-1\right)^2-7\left(\frac{1}{x}-1\right)-4=0$$

Note 0.1.5, $t=\frac{1}{x}-1$.

$$\begin{aligned}2t^2-7t-4&=0\\2t^2-8t+t-4&=0\\(2t+1)(t-4)&=0\\t&\in\left\{-\frac{1}{2},4\right\}\end{aligned}$$

Note 0.1.5, $t=-\frac{1}{2}$.

$$\frac{1}{x}-1=-\frac{1}{2}$$

$x\neq 0$

$$\begin{aligned}2-2x&=-x\\x&=2\end{aligned}$$

Note 0.1.5, $t=4$.

$$\frac{1}{x}-1=4$$

$x\neq 0$

$$\begin{aligned}1-x&=4x\\5x&=1\\x&=\frac{1}{5}\end{aligned}$$

$$\boxed{x\in\left\{\frac{1}{5},2\right\}}$$

1.49

$$x^2(x^2-1)-9(x^2-1)=0$$

$$\begin{aligned}(x^2-9)(x^2-1)&=0\\(x-3)(x+3)(x-1)(x+1)&=0\end{aligned}$$

$$\boxed{x \in \{\pm 1, \pm 3\}}$$

1.50

$$2(x^2+x+1)+\sqrt{x^2+x+1}-3=0$$

Note 0.1.5, $t=\sqrt{x^2+x+1}$.

$$\begin{aligned}2t^2+t-3&=0\\2t^2+3t-2t-3&=0\\(t-1)(2t+3)&=0\\t&\in\left\{-\frac{3}{2},1\right\}\end{aligned}$$

Note 0.1.5, $t=-\frac{3}{2}$.

$$\sqrt{x^2+x+1}=-\frac{3}{2}$$

Note 0.2.1

$$x\in\emptyset$$

Note 0.1.5, $t=1$.

$$\begin{aligned}\sqrt{x^2+x+1}&=1\\x^2+x+1&=1\\x^2+x&=0\\x(x+1)&=0\\x&\in\{-1,0\}\end{aligned}$$

Note 0.2.2, $x=-1$.

$$\begin{aligned}\sqrt{(-1)^2-1+1}&=1\\1&=1\end{aligned}$$

Note 0.2.2, $x=0$.

$$\begin{aligned}\sqrt{0^2-0+1}&=1\\1&=1\end{aligned}$$

$$\boxed{x\in\{-1,0\}}$$

1.51

$$|7 - 2w| = 9$$

Note 0.3.1, positive case.

$$\begin{aligned} 7 - 2w &= 9 \\ 2w &= -2 \\ w &= -1 \end{aligned}$$

Note 0.3.1, negative case.

$$\begin{aligned} 7 - 2w &= -9 \\ 2w &= 16 \\ w &= 8 \end{aligned}$$

$$\boxed{w \in \{-1, 8\}}$$

1.52

$$|3x - 2| = |2x + 3|$$

Note 0.3.1, positive case.

$$\begin{aligned} 3x - 2 &= 2x + 3 \\ x &= 5 \end{aligned}$$

Note 0.3.1, negative case.

$$\begin{aligned} 3x - 2 &= -2x - 3 \\ 5x &= -1 \\ x &= -\frac{1}{5} \end{aligned}$$

$$\boxed{x \in \left\{-\frac{1}{5}, 5\right\}}$$

1.53

$$|x-4| = |5-2x|$$

Note 0.3.1, positive case.

$$\begin{aligned} x-4 &= 5-2x \\ 3x &= 9 \\ x &= 3 \end{aligned}$$

Note 0.3.1, negative case.

$$\begin{aligned} x-4 &= -5+2x \\ x &= 1 \end{aligned}$$

$$\boxed{x \in \{1,3\}}$$

1.54

$$\left|\frac{x+3}{x-3}\right| = 7$$

Note 0.3.1, positive case.

$$\frac{x+3}{x-3} = 7$$

$x \neq 3$

$$\begin{aligned} x+3 &= 7x-21 \\ 6x &= 24 \\ x &= 4 \end{aligned}$$

Note 0.3.1, negative case.

$$\frac{x+3}{x-3} = -7$$

$x \neq 3$

$$\begin{aligned} x+3 &= -7x+21 \\ 8x &= 18 \\ x &= \frac{18}{8} \\ x &= \frac{9}{4} \end{aligned}$$

$$\boxed{x \in \left\{\frac{9}{4}, 4\right\}}$$

1.55

$$\left|\frac{2x+1}{x-1}\right| = 3$$

Note 0.3.1, positive case.

$$\frac{2x+1}{x-1} = 3$$

$x \neq 1$

$$\begin{aligned} 2x+1 &= 3x-3 \\ x &= 4 \end{aligned}$$

Note 0.3.1, negative case.

$$\frac{2x+1}{x-1} = -3$$

$x \neq 1$

$$\begin{aligned} 2x+1 &= -3x+3 \\ 5x &= 2 \\ x &= \frac{2}{5} \end{aligned}$$

$$\boxed{x \in \left\{\frac{2}{5}, 4\right\}}$$

1.56

$$|3t-2| = t^2$$

Note 0.3.1, $t \geq \frac{2}{3}$.

$$\begin{aligned} 3t-2 &= t^2 \\ t^2-3t+2 &= 0 \\ (t-2)(t-1) &= 0 \\ t &\in \{1,2\} \end{aligned}$$

Note 0.3.1, $t < \frac{2}{3}$.

$$\begin{aligned} 3t-2 &= -t^2 \\ t^2+3t-2 &= 0 \end{aligned}$$

Note 0.1.3

$$\begin{aligned} t &= \frac{-3 \pm \sqrt{3^2-4(1)(-2)}}{2(1)} \\ t &= \frac{-3 \pm \sqrt{9+8}}{2} \\ t &= \frac{-3 \pm \sqrt{17}}{2} \end{aligned}$$

$$\boxed{t \in \left\{1, 2, \frac{-3 \pm \sqrt{17}}{2}\right\}}$$

1.57

$$|3t-2| = -t^2$$

Note 0.3.1, $t \geq \frac{2}{3}$.

$$\begin{aligned} 3t-2 &= -t^2 \\ t^2+3t-2 &= 0 \end{aligned}$$

Note 0.1.3

$$\begin{aligned} t &= \frac{-3 \pm \sqrt{3^2-4(1)(-2)}}{2(1)} \\ t &= \frac{-3 \pm \sqrt{9+8}}{2} \\ t &= \frac{-3 \pm \sqrt{17}}{2} \\ t &\in \emptyset \end{aligned}$$

Note 0.3.1, $t < \frac{2}{3}$.

$$\begin{aligned} 3t-2 &= t^2 \\ t^2-3t+2 &= 0 \\ (t-2)(t-1) &= 0 \\ t &\in \{1,2\} \\ t &\in \emptyset \end{aligned}$$

$$\boxed{t \in \emptyset}$$

1.58

$$|3x+5| = 9$$

Note 0.3.1, positive case.

$$\begin{aligned} 3x+5 &= 9 \\ 3x &= 4 \\ x &= \frac{4}{3} \end{aligned}$$

Note 0.3.1, negative case.

$$\begin{aligned} 3x+5 &= -9 \\ 3x &= -14 \\ x &= -\frac{14}{3} \end{aligned}$$

$$\boxed{x \in \left\{-\frac{14}{3}, \frac{4}{3}\right\}}$$

1.59

$$|-3x+7|-3=0$$

$$|-3x+7|=3$$

Note 0.3.1, positive case.

$$\begin{aligned} -3x+7&=3 \\ 3x&=4 \\ x&=\frac{4}{3} \end{aligned}$$

Note 0.3.1, negative case.

$$\begin{aligned} -3x+7&=-3 \\ 3x&=10 \\ x&=\frac{10}{3} \end{aligned}$$

$$\boxed{x \in \left\{\frac{4}{3}, \frac{10}{3}\right\}}$$

1.60

$$\left|\frac{x}{2x-3}\right| = 2$$

Note 0.3.1, positive case.

$$\frac{x}{2x-3} = 2$$

$x \neq \frac{3}{2}$

$$\begin{aligned} x &= 4x - 6 \\ 3x &= 6 \\ x &= 2 \end{aligned}$$

Note 0.3.1, negative case.

$$\frac{x}{2x-3} = -2$$

$x \neq \frac{3}{2}$

$$\begin{aligned} x &= -4x + 6 \\ 5x &= 6 \\ x &= \frac{6}{5} \end{aligned}$$

$$\boxed{x \in \left\{\frac{6}{5}, 2\right\}}$$

1.61

$$\left|\frac{3x-4}{2x+3}\right| = 1$$

Note 0.3.1, positive case.

$$\frac{3x-4}{2x+3} = 1$$

$x \neq -\frac{3}{2}$

$$\begin{aligned} 3x - 4 &= 2x + 3 \\ x &= 7 \end{aligned}$$

Note 0.3.1, negative case.

$$\frac{3x-4}{2x+3} = -1$$

$x \neq -\frac{3}{2}$

$$\begin{aligned} 3x - 4 &= -2x - 3 \\ 5x &= 1 \\ x &= \frac{1}{5} \end{aligned}$$

$$\boxed{x \in \left\{\frac{1}{5}, 7\right\}}$$

1.62

$$|2x - 3| = |7 - 3x|$$

Note 0.3.1, positive case.

$$\begin{aligned} 2x - 3 &= 7 - 3x \\ 5x &= 10 \\ x &= 2 \end{aligned}$$

Note 0.3.1, negative case.

$$\begin{aligned} 2x - 3 &= -7 + 3x \\ x &= 4 \end{aligned}$$

$$\boxed{x \in \{2, 4\}}$$

1.63

$$|4x - 3| = x$$

Note 0.3.1, $x \geq \frac{3}{4}$.

$$\begin{aligned} 4x - 3 &= x \\ 3x &= 3 \\ x &= 1 \end{aligned}$$

Note 0.3.1, $x < \frac{3}{4}$.

$$\begin{aligned} 4x - 3 &= -x \\ 5x &= 3 \\ x &= \frac{3}{5} \end{aligned}$$

$$\boxed{x \in \left\{\frac{3}{5}, 1\right\}}$$

1.64

$$|3x+8| = x$$

Note 0.3.1, $x \geq -\frac{8}{3}$.

$$\begin{aligned} 3x+8 &= x \\ 2x &= -8 \\ x &= -4 \\ x &\in \emptyset \end{aligned}$$

Note 0.3.1, $x < -\frac{8}{3}$.

$$\begin{aligned} 3x+8 &= -x \\ 4x &= -8 \\ x &= -2 \\ x &\in \emptyset \end{aligned}$$

$$\boxed{x \in \emptyset}$$

1.65

$$|4x - 3| = -x$$

Note 0.3.1, $x \geq \frac{3}{4}$.

$$\begin{aligned} 4x - 3 &= -x \\ 5x &= 3 \\ x &= \frac{3}{5} \\ x &\in \emptyset \end{aligned}$$

Note 0.3.1, $x < \frac{3}{4}$.

$$\begin{aligned} 4x - 3 &= x \\ 3x &= 3 \\ x &= 1 \\ x &\in \emptyset \end{aligned}$$

$$\boxed{x \in \emptyset}$$

1.66

$$|3x+8| = -x$$

Note 0.3.1, $x \geq -\frac{8}{3}$.

$$\begin{aligned} 3x+8 &= -x \\ 4x &= -8 \\ x &= -2 \end{aligned}$$

Note 0.3.1, $x < -\frac{8}{3}$.

$$\begin{aligned} 3x+8 &= x \\ 2x &= -8 \\ x &= -4 \end{aligned}$$

$$\boxed{x \in \{-4, -2\}}$$

2

2.1

Find all real values of k such that there is exactly one solution to the equation

$$x^2 - kx + k = x - 2$$

$$x^2 - (k+1)x + (k+2) = 0$$

Note 0.1.2, we want the discriminant to be zero for it to have only one solution.

$$\begin{aligned}(k+1)^2 - 4(1)(k+2) &= 0\\ k^2 + 2k + 1 - 4k - 8 &= 0\\ k^2 - 2k - 7 &= 0\end{aligned}$$

Note 0.1.3

$$\begin{aligned}k &= \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-7)}}{2(1)}\\ k &= \frac{2 \pm \sqrt{4+28}}{2}\\ k &= \frac{2 \pm 4\sqrt{2}}{2}\end{aligned}$$

$$\boxed{k = 1 \pm 2\sqrt{2}}$$

3

Find the solution set of the following inequalities.

3.1

$$x - 2 < 4 - x$$

$$2x < 6$$
$$x < 3$$

$$\boxed{x \in (-\infty, 3)}$$

3.2

$$x + 1 > \frac{2x - 4}{5}$$

$$5x + 5 > 2x - 4$$
$$3x > -9$$
$$x > -3$$

$$\boxed{x \in (-3, +\infty)}$$

3.3

$$\frac{2x-3}{4} \geq \frac{7x}{2}$$

$$\begin{aligned} 2x-3 &\geq 14x \\ 12x &\leq -3 \\ x &\leq -\frac{3}{12} \\ x &\leq -\frac{1}{4} \end{aligned}$$

$$\boxed{x \in \left(-\infty, -\frac{1}{4}\right]}$$

3.4

$$-2 < 5 + 3x < 20$$

$$-7 < 3x < 15$$

$$-\frac{7}{3} < x < 5$$

$$\boxed{x \in \left(-\frac{7}{3}, 5\right)}$$

3.5

$$5 \leq 3x - 4 \leq 7$$

$$9 \leq 3x \leq 11$$
$$3 \leq x \leq \frac{11}{3}$$

$$\boxed{x \in \left[3, \frac{11}{3}\right]}$$

3.6

$$2x - 1 \leq 3x \leq 4x + 1$$

Solve left inequality.

$$\begin{aligned} 2x - 1 &\leq 3x \\ x &\geq 1 \\ x &\in [1, +\infty) \end{aligned}$$

Solve right inequality.

$$\begin{aligned} 3x &\leq 4x + 1 \\ x &\geq -1 \\ x &\in [-1, +\infty) \end{aligned}$$

Intersect intervals.

$$\boxed{x \in [-1, +\infty)}$$

3.7

$$3x^2 \leq 11x - 10$$

$$\begin{aligned} 3x^2 - 11x + 10 &\leq 0 \\ 3x^2 - 6x - 5x + 10 &\leq 0 \\ (3x-5)(x-2) &\leq 0 \end{aligned}$$

Note 0.4.1, $c \in \left\{\frac{5}{3}, 2\right\}$.

		$\frac{5}{3}$		2	
$3x - 5$	$-$		$+$		$+$
$x - 2$	$-$		$-$		$+$
$(3x-5)(x-2)$	$+$		$-$		$+$

$$\boxed{x \in \left[\frac{5}{3}, 2\right]}$$

3.8

$$(x+3)(x-4) \geq 3x$$

$$\begin{aligned} x^2 - 4x + 3x - 12 &\geq 3x \\ x^2 - 4x - 12 &\geq 0 \\ (x-6)(x+2) &\geq 0 \end{aligned}$$

Note 0.4.1, $c \in \{-2, 6\}$.

		-2		6	
$x - 6$	$-$		$-$		$+$
$x + 2$	$-$		$+$		$+$
$(x-6)(x+2)$	$+$		$-$		$+$

$$\boxed{x \in (-\infty, -2] \cup [6, +\infty)}$$

3.9

$$x^3 + x \geq x^2 + 1$$

$$x^3 - x^2 + x - 1 \geq 0$$
$$(x^2 + 1)(x - 1) \geq 0$$

$x^2 + 1$ is never negative, so only consider $x - 1$.

$$x - 1 \geq 0$$
$$x \geq 1$$

$$\boxed{x \in [1, +\infty)}$$

3.10

$$x^2 > 9$$

$$\begin{aligned} x^2 - 9 &> 0 \\ (x-3)(x+3) &> 0 \end{aligned}$$

Note 0.4.1, $c \in \{-3, 3\}$.

		-3		3	
$x - 3$	$-$		$-$		$+$
$x + 3$	$-$		$+$		$+$
$(x-3)(x+3)$	$+$		$-$		$+$

$$\boxed{x \in (-\infty, -3) \cup (3, +\infty)}$$

3.11

$$4 - 3x - x^2 \geq 0$$

$$\begin{aligned} x^2 + 3x - 4 &\leq 0 \\ (x+4)(x-1) &\leq 0 \end{aligned}$$

Note 0.4.1, $c \in \{-4, 1\}$.

		-4		1	
$x + 4$	$-$		$+$		$+$
$x - 1$	$-$		$-$		$+$
$(x+4)(x-1)$	$+$		$-$		$+$

$$\boxed{x \in [-4, 1]}$$

3.12

$$4x^2 + 4x + 1 < 0$$

$$(2x + 1)^2 < 0$$

Note 0.1.1

$$\boxed{x \in \emptyset}$$

3.13

$$4x^2 + 4x > -1$$

$$\begin{aligned} 4x^2 + 4x + 1 &> 0 \\ (2x+1)^2 &> 0 \end{aligned}$$

This is always true unless $x = -\frac{1}{2}$.

$$\boxed{x \in \left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, +\infty\right)}$$

3.14

$$x^2 - 2x < 2x - 3$$

$$\begin{aligned} x^2 - 4x + 3 &< 0 \\ (x-3)(x-1) &< 0 \end{aligned}$$

Note 0.4.1, $c \in \{1, 3\}$.

		1		3	
$x - 3$	$-$		$-$		$+$
$x - 1$	$-$		$+$		$+$
$(x-3)(x-1)$	$+$		$-$		$+$

$$\boxed{x \in (1, 3)}$$

3.15

$$(x+4)(x^2-4) < 0$$

$$(x+4)(x-2)(x+2) < 0$$

Note 0.4.1, $c \in \{-4, -2, 2\}$.

		-4		-2		2	
$x + 4$	$-$		$+$		$+$		$+$
$x - 2$	$-$		$-$		$-$		$+$
$x + 2$	$-$		$-$		$+$		$+$
$(x+4)(x-2)(x+2)$	$-$		$+$		$-$		$+$

$$\boxed{x \in (-\infty, -4) \cup (-2, 2)}$$

3.16

$$x - 2 \geq (x-2)(2x^2 - x - 14)$$

$$\begin{aligned}
(x-2) - (x-2)(2x^2 - x - 14) &\geq 0 \\
(x-2)(1 - 2x^2 + x + 14) &\geq 0 \\
(x-2)(-2x^2 + x + 15) &\geq 0 \\
(x-2)(2x^2 - x - 15) &\leq 0 \\
(x-2)(2x^2 - 6x + 5x - 15) &\leq 0 \\
(x-2)(2x+5)(x-3) &\leq 0
\end{aligned}$$

Note 0.4.1, $c \in \left\{-\frac{5}{2}, 2, 3\right\}$.

		$-\frac{5}{2}$		2		3	
$x - 2$	$-$		$-$		$+$		$+$
$2x + 5$	$-$		$+$		$+$		$+$
$x - 3$	$-$		$-$		$-$		$+$
$(x-2)(2x+5)(x-3)$	$-$		$+$		$-$		$+$

$$\boxed{x \in \left(-\infty, -\frac{5}{2}\right] \cup [2, 3]}$$

3.17

$$x(2x^2 - 5x - 12) \geq 2x^2 - 5x - 12$$

$$\begin{aligned} x(2x^2 - 5x - 12) - (2x^2 - 5x - 12) &\geq 0 \\ (x-1)(2x^2 - 5x - 12) &\geq 0 \\ (x-1)(2x^2 - 8x + 3x - 12) &\geq 0 \\ (x-1)(2x+3)(x-4) &\geq 0 \end{aligned}$$

Note 0.4.1, $c \in \left\{-\frac{3}{2}, 1, 4\right\}$.

		$-\frac{3}{2}$		1		4	
$x - 1$	$-$		$-$		$+$		$+$
$2x + 3$	$-$		$+$		$+$		$+$
$x - 4$	$-$		$-$		$-$		$+$
$(x-1)(2x+3)(x-4)$	$-$		$+$		$-$		$+$

$$\boxed{x \in \left[-\frac{3}{2}, 1\right] \cup [4, +\infty)}$$

3.18

$$\frac{x}{x-1} > -1$$

$$\frac{x+x-1}{x-1} > 0$$
$$\frac{2x-1}{x-1} > 0$$

Note 0.4.1, $c \in \left\{\frac{1}{2}, 1\right\}$.

		$\frac{1}{2}$		1	
$2x-1$	$-$		$+$		$+$
$x-1$	$-$		$-$		$+$
$\frac{2x-1}{x-1}$	$+$		$-$		$+$

$$\boxed{x \in \left(-\infty, \frac{1}{2}\right) \cup (1, +\infty)}$$

3.19

$$\frac{x-1}{2x+5} \geq 0$$

Note 0.4.1, $c \in \left\{-\frac{5}{2}, 1\right\}$, $x \neq -\frac{5}{2}$.

		$-\frac{5}{2}$		1	
$x-1$	$-$		$-$		$+$
$2x+5$	$-$		$+$		$+$
$\frac{x-1}{2x+5}$	$+$	$\Vert$	$-$		$+$

$$\boxed{x \in \left(-\infty, -\frac{5}{2}\right) \cup [1, +\infty)}$$

3.20

$$\frac{6-2x}{4+x} > 5$$

$$\frac{6-2x-5x-20}{x+4} > 0$$
$$\frac{-7x-14}{x+4} > 0$$
$$\frac{x+2}{x+4} < 0$$

Note 0.4.1, $c \in \{-4, -2\}$.

		-4		-2	
$x+2$	$-$		$-$		$+$
$x+4$	$-$		$+$		$+$
$\frac{x+2}{x+4}$	$+$		$-$		$+$

$$\boxed{x \in (-4, -2)}$$

3.21

$$\frac{x^2 - x - 2}{x} \leq 0$$

$$\frac{(x-2)(x+1)}{x} \leq 0$$

Note 0.4.1, $c \in \{-1, 0, 2\}$, $x \neq 0$.

		-1		0		2	
$x - 2$	$-$		$-$		$-$		$+$
$x + 1$	$-$		$+$		$+$		$+$
x	$-$		$-$		$+$		$+$
$\frac{(x-2)(x+1)}{x}$	$-$		$+$	$\Vert$	$-$		$+$

$$\boxed{x \in (-\infty, -1] \cup (0, 2]}$$

3.22

$$\frac{3x-1}{x^2-x-6} \leq 1$$

$$\begin{aligned}
\frac{3x-1-x^2+x+6}{x^2-x-6} &\leq 0 \\
\frac{-x^2+4x+5}{x^2-x-6} &\leq 0 \\
\frac{x^2-4x-5}{x^2-x-6} &\geq 0 \\
\frac{(x-5)(x+1)}{(x-3)(x+2)} &\geq 0
\end{aligned}$$

Note 0.4.1, $c \in \{-2, -1, 3, 5\}$, $x \notin \{-2, 3\}$.

		-2		-1		3		5	
$x-5$	$-$		$-$		$-$		$-$		$+$
$x+1$	$-$		$-$		$+$		$+$		$+$
$x-3$	$-$		$-$		$-$		$+$		$+$
$x+2$	$-$		$+$		$+$		$+$		$+$
$\frac{(x-5)(x+1)}{(x-3)(x+2)}$	$+$	$\Vert$	$-$		$+$	$\Vert$	$-$		$+$

$$\boxed{x \in (-\infty, -2) \cup [-1, 3) \cup [5, +\infty)}$$

3.23

$$\frac{x}{x-2} \leq 1$$

$$\begin{aligned}
\frac{x-x+2}{x-2} &\leq 0 \\
\frac{2}{x-2} &\leq 0 \\
x-2 &< 0 \\
x &< 2
\end{aligned}$$

$$\boxed{x \in (-\infty, 2)}$$

3.24

$$\frac{1}{x-2} \leq \frac{1}{x}$$

$$\begin{aligned}\frac{x-x+2}{x(x-2)} &\leq 0\\ \frac{2}{x(x-2)} &\leq 0\\ x(x-2) &< 0\end{aligned}$$

Note 0.4.1, $c \in \{0, 2\}$.

		0		2	
$x-2$	$-$		$-$		$+$
x	$-$		$+$		$+$
$x(x-2)$	$+$		$-$		$+$

$$\boxed{x \in (0, 2)}$$

3.25

$$\frac{1}{x-2} \leq \frac{1}{3}$$

$$\begin{aligned}
\frac{3}{x-2} &\leq 1 \\
\frac{3-x+2}{x-2} &\leq 0 \\
\frac{-x+5}{x-2} &\leq 0 \\
\frac{x-5}{x-2} &\geq 0
\end{aligned}$$

Note 0.4.1, $c \in \{2, 5\}$, $x \neq 2$.

		2		5	
$x-5$	$-$		$-$		$+$
$x-2$	$-$		$+$		$+$
$\frac{x-5}{x-2}$	$+$	$\Vert$	$-$		$+$

$$\boxed{x \in (-\infty, 2) \cup [5, +\infty)}$$

3.26

$$\frac{x+1}{x-7} \geq 2 + \frac{1}{x-7}$$

$$\begin{aligned} \frac{x+1-2x+14-1}{x-7} &\geq 0 \\ \frac{-x+14}{x-7} &\geq 0 \\ \frac{x-14}{x-7} &\leq 0 \end{aligned}$$

Note 0.4.1, $c \in \{7, 14\}$, $x \neq 7$.

		7		14	
$x-14$	$-$		$-$		$+$
$x-7$	$-$		$+$		$+$
$\frac{x-14}{x-7}$	$+$	$\Vert$	$-$		$+$

$$\boxed{x \in (7, 14]}$$

3.27

$$\frac{x^2}{x+1} \geq -5 - \frac{1}{x+1}$$

$$\frac{x^2+5x+5+1}{x+1} \geq 0$$
$$\frac{x^2+5x+6}{x+1} \geq 0$$
$$\frac{(x+3)(x+2)}{x+1} \geq 0$$

Note 0.4.1, $c \in \{-3, -2, -1\}$, $x \neq 1$.

		-3		-2		-1	
$x+3$	$-$		$+$		$+$		$+$
$x+2$	$-$		$-$		$+$		$+$
$x+1$	$-$		$-$		$-$		$+$
$\frac{(x+3)(x+2)}{x+1}$	$-$		$+$		$-$	$\Vert$	$+$

$$\boxed{x \in [-3, -2] \cup (-1, +\infty)}$$

3.28

$$\frac{x}{(2-x)(x+1)} \geq 1$$

$$\begin{aligned}
\frac{x-(2-x)(x+1)}{(2-x)(x+1)} &\geq 0 \\
\frac{x-2x-2+x^2+x}{(2-x)(x+1)} &\geq 0 \\
\frac{x^2-2}{(2-x)(x+1)} &\geq 0 \\
\frac{(x-\sqrt{2})(x+\sqrt{2})}{(x-2)(x+1)} &\leq 0
\end{aligned}$$

Note 0.4.1, $c \in \{\pm\sqrt{2}, -1, 2\}$, $x \notin \{-1, 2\}$.

		$-\sqrt{2}$		-1		$\sqrt{2}$		2	
$x - \sqrt{2}$	$-$		$-$		$-$		$+$		$+$
$x + \sqrt{2}$	$-$		$+$		$+$		$+$		$+$
$x - 2$	$-$		$-$		$-$		$-$		$+$
$x + 1$	$-$		$-$		$+$		$+$		$+$
$\frac{(x-\sqrt{2})(x+\sqrt{2})}{(x-2)(x+1)}$	$+$		$-$	$\Vert$	$+$		$-$	$\Vert$	$+$

$$\boxed{x \in [-\sqrt{2}, -1) \cup [\sqrt{2}, 2)}$$

3.29

$$\frac{2x-7}{x^2-6x+8} \leq 1$$

$$\begin{aligned}
\frac{2x-7-x^2+6x-8}{x^2-6x+8} &\leq 0 \\
\frac{-x^2+8x-15}{x^2-6x+8} &\leq 0 \\
\frac{x^2-8x+15}{x^2-6x+8} &\geq 0 \\
\frac{(x-5)(x-3)}{(x-4)(x-2)} &\geq 0
\end{aligned}$$

Note 0.4.1, $c \in \{2,3,4,5\}$, $x \notin \{2,4\}$.

		2		3		4		5	
$x-5$	$-$		$-$		$-$		$-$		$+$
$x-3$	$-$		$-$		$+$		$+$		$+$
$x-4$	$-$		$-$		$-$		$+$		$+$
$x-2$	$-$		$+$		$+$		$+$		$+$
$\frac{(x-5)(x-3)}{(x-4)(x-2)}$	$+$	‖	$-$		$+$	‖	$-$		$+$

$$\boxed{x \in (-\infty, 2) \cup [3,4) \cup [5,+\infty)}$$

3.30

$$\frac{x}{x+1} \geq \frac{2}{x+3}$$

$$\frac{x^2+3x-2x-2}{(x+1)(x+3)} \geq 0$$
$$\frac{x^2+x-2}{(x+1)(x+3)} \geq 0$$
$$\frac{(x+2)(x-1)}{(x+1)(x+3)} \geq 0$$

Note 0.4.1, $c \in \{-3,-2,-1,1\}$, $x \notin \{-3,-1\}$.

		-3		-2		-1		1	
$x+2$	$-$		$-$		$+$		$+$		$+$
$x-1$	$-$		$-$		$-$		$-$		$+$
$x+1$	$-$		$-$		$-$		$+$		$+$
$x+3$	$-$		$+$		$+$		$+$		$+$
$\frac{(x+2)(x-1)}{(x+1)(x+3)}$	$+$	$\Vert$	$-$		$+$	$\Vert$	$-$		$+$

$$\boxed{x \in (-\infty,-3) \cup [-2,-1) \cup [1,+\infty)}$$

3.31

$$\frac{4}{2x-1} < \frac{1}{x+1}$$

$$\frac{4x+4-2x+1}{(2x-1)(x+1)} < 0$$
$$\frac{2x+5}{(2x-1)(x+1)} < 0$$

Note 0.4.1, $c \in \left\{-\frac{5}{2}, -1, \frac{1}{2}\right\}$.

		$-\frac{5}{2}$		-1		$\frac{1}{2}$	
$2x+5$	$-$		$+$		$+$		$+$
$2x-1$	$-$		$-$		$-$		$+$
$x+1$	$-$		$-$		$+$		$+$
$\frac{2x+5}{(2x-1)(x+1)}$	$-$		$+$		$-$		$+$

$$\boxed{x \in \left(-\infty, -\frac{5}{2}\right) \cup \left(-1, \frac{1}{2}\right)}$$

3.32

$$\frac{6}{3w-4} \geq w+1$$

$$\begin{aligned}
\frac{6-3w^2+4w-3w+4}{3w-4} &\geq 0 \\
\frac{-3w^2+w+10}{3w-4} &\geq 0 \\
\frac{3w^2-w-10}{3w-4} &\leq 0 \\
\frac{3w^2-6w+5w-10}{3w-4} &\leq 0 \\
\frac{(3w+5)(w-2)}{3w-4} &\leq 0
\end{aligned}$$

Note 0.4.1, $c \in \left\{-\frac{5}{3}, \frac{4}{3}, 2\right\}$, $w \neq \frac{4}{3}$.

		$-\frac{5}{3}$		$\frac{4}{3}$		2	
$3w+5$	$-$		$+$		$+$		$+$
$w-2$	$-$		$-$		$-$		$+$
$3w-4$	$-$		$-$		$+$		$+$
$\frac{(3w+5)(w-2)}{3w-4}$	$-$		$+$	$\Vert$	$-$		$+$

$$\boxed{w \in \left(-\infty, -\frac{5}{3}\right] \cup \left(\frac{4}{3}, 2\right]}$$

3.33

$$\frac{2x+1}{4} \leq \frac{2x}{3} + \frac{1}{6}$$

$$6x + 3 \leq 8x + 2$$
$$2x \geq 1$$
$$x \geq \frac{1}{2}$$

$$\boxed{x \in \left[\frac{1}{2}, +\infty\right)}$$

3.34

$$|x+2| \geq 3$$

Note 0.4.2, positive case.

$$\begin{aligned} x+2 &\geq 3 \\ x &\geq 1 \end{aligned}$$

Note 0.4.2, negative case.

$$\begin{aligned} x+2 &\leq -3 \\ x &\leq -5 \end{aligned}$$

Note 0.4.2, unite intervals.

$$\boxed{x \in (-\infty, -5] \cup [1, +\infty)}$$

3.35

$$|3-2x| > 2$$

Note 0.4.2, positive case.

$$\begin{aligned} 3-2x &> 2 \\ 2x &< 1 \\ x &< \frac{1}{2} \end{aligned}$$

Note 0.4.2, negative case.

$$\begin{aligned} 3-2x &< -2 \\ 2x &> 5 \\ x &> \frac{5}{2} \end{aligned}$$

Note 0.4.2, unite intervals.

$$\boxed{x \in \left(-\infty, \frac{1}{2}\right) \cup \left(\frac{5}{2}, +\infty\right)}$$

3.36

$$|3 + 2x| + 8 \leq 7$$

$$|3 + 2x| \leq -1$$

Note 0.3.1

$$\boxed{x \in \emptyset}$$

3.37

$$|3x - 1| + 2 \geq 1$$

$$|3x - 1| \geq -1$$

Note 0.3.1

$$\boxed{x \in \mathbb{R}}$$

3.38

$$|x^2 - 6x + 7| \geq 2$$

Note 0.4.2, positive case.

$$\begin{aligned} x^2 - 6x + 7 &\geq 2 \\ x^2 - 6x + 5 &\geq 0 \\ (x-5)(x-1) &\geq 0 \end{aligned}$$

Note 0.4.1, $c \in \{1, 5\}$.

		1		5	
$x - 5$	$-$		$-$		$+$
$x - 1$	$-$		$+$		$+$
$(x-5)(x-1)$	$+$		$-$		$+$

$$x \in (-\infty, 1] \cup [5, +\infty)$$

Note 0.4.2, negative case.

$$\begin{aligned} x^2 - 6x + 7 &\leq -2 \\ x^2 - 6x + 9 &\leq 0 \\ (x-3)^2 &\leq 0 \end{aligned}$$

This is only true if $x = 3$. Note 0.4.2, unite intervals.

$$\boxed{x \in (-\infty, 1] \cup \{3\} \cup [5, +\infty)}$$

3.39

$$\left|\frac{3x+9}{2x-2}\right| < -3$$

Note 0.3.1

$$\boxed{x \in \emptyset}$$

3.40

$$\left|\frac{4}{x}+1\right| \geq 3$$

Note 0.4.2, positive case.

$$\begin{aligned}\frac{4}{x}+1 &\geq 3\\ \frac{4+x-3x}{x} &\geq 0\\ \frac{4-2x}{x} &\geq 0\\ \frac{x-2}{x} &\leq 0\end{aligned}$$

Note 0.4.1, $c \in \{0, 2\}$, $x \neq 0$.

		0		2	
$x-2$	$-$		$-$		$+$
x	$-$		$+$		$+$
$\frac{x-2}{x}$	$+$	$\Vert$	$-$		$+$

$x \in (0, 2]$

Note 0.4.2, negative case.

$$\begin{aligned}\frac{4}{x}+1 &\leq -3\\ \frac{4+x+3x}{x} &\leq 0\\ \frac{4x+4}{x} &\leq 0\\ \frac{x+1}{x} &\leq 0\end{aligned}$$

Note 0.4.1, $c \in \{-1, 0\}$, $x \neq 0$.

		-1		0	
$x+1$	$-$		$+$		$+$
x	$-$		$-$		$+$
$\frac{x+1}{x}$	$+$		$-$	$\Vert$	$+$

$x \in [-1, 0)$

Note 0.4.2, unite intervals.

$$\boxed{x \in [-1, 0) \cup (0, 2]}$$

3.41

$$\left|\frac{x}{2x-3}\right| \leq 1$$

Note 0.4.2, positive case.

$$\begin{aligned}
\frac{x}{2x-3} &\leq 1 \\
\frac{x-2x+3}{2x-3} &\leq 0 \\
\frac{-x+3}{2x-3} &\leq 0 \\
\frac{x-3}{2x-3} &\geq 0
\end{aligned}$$

Note 0.4.1, $c \in \left\{\frac{3}{2}, 3\right\}$, $x \neq \frac{3}{2}$.

		$\frac{3}{2}$		3	
$x-3$	$-$		$-$		$+$
$2x-3$	$-$		$+$		$+$
$\frac{x-3}{2x-3}$	$+$	$\Vert$	$-$		$+$

$$x \in \left(-\infty, \frac{3}{2}\right) \cup [3, +\infty)$$

Note 0.4.2, negative case.

$$\begin{aligned}
\frac{x}{2x-3} &\geq -1 \\
\frac{x+2x-3}{2x-3} &\geq 0 \\
\frac{3x-3}{2x-3} &\geq 0 \\
\frac{x-1}{2x-3} &\geq 0
\end{aligned}$$

Note 0.4.1, $c \in \left\{1, \frac{3}{2}\right\}$, $x \neq \frac{3}{2}$.

		1		$\frac{3}{2}$	
$x-1$	$-$		$+$		$+$
$2x-3$	$-$		$-$		$+$
$\frac{x-1}{2x-3}$	$+$		$-$	$\Vert$	$+$

$$x \in (-\infty, 1] \cup \left(\frac{3}{2}, +\infty\right)$$

Note 0.4.2, intersect intervals.

$$\boxed{x \in (-\infty, 1] \cup [3, +\infty)}$$

3.42

$$0 < |x - 5| < 2$$

Solve left inequality.

$$|x - 5| > 0$$

This is always true unless $x = 5$.

$$x \in (-\infty, 5] \cup [5, +\infty)$$

Solve right inequality.

$$|x - 5| < 2$$

Note 0.4.2, positive case.

$$\begin{aligned} x - 5 &< 2 \\ x &< 7 \\ x &\in (-\infty, 7) \end{aligned}$$

Note 0.4.2, negative case.

$$\begin{aligned} x - 5 &> -2 \\ x &> 3 \\ x &\in (3, +\infty) \end{aligned}$$

Note 0.4.2, intersect intervals.

$$x \in (3, 7)$$

Intersect intervals.

$$\boxed{x \in (3, 5) \cup (5, 7)}$$

3.43

$$3|8-x|+2<7-2|x-8|$$

Note 0.3.2

$$\begin{aligned}3|-1||x-8|+2|x-8|&<5\\3|x-8|+2|x-8|&<5\\5|x-8|&<5\\|x-8|&<1\end{aligned}$$

Note 0.4.2, positive case.

$$\begin{aligned}x-8&<1\\x&<9\\x&\in(-\infty,9)\end{aligned}$$

Note 0.4.2, negative case.

$$\begin{aligned}x-8&>-1\\x&>7\\x&\in(7,+\infty)\end{aligned}$$

Note 0.4.2, intersect intervals.

$$\boxed{x\in(7,9)}$$

3.44

$$\left|\frac{9-2x}{4x}\right| \geq 1$$

Note 0.3.2

$$\begin{aligned} |-1|\left|\frac{2x-9}{4x}\right| &\geq 1 \\ \left|\frac{2x-9}{4x}\right| &\geq 1 \end{aligned}$$

Note 0.4.2, positive case.

$$\begin{aligned} \frac{2x-9}{4x} &\geq 1 \\ \frac{2x-9-4x}{4x} &\geq 0 \\ \frac{-2x-9}{4x} &\geq 0 \\ \frac{2x+9}{4x} &\leq 0 \end{aligned}$$

Note 0.4.1, $c \in \left\{-\frac{9}{2}, 0\right\}$, $x \neq 0$.

		$-\frac{9}{2}$		0	
$2x+9$	$-$		$+$		$+$
$4x$	$-$		$-$		$+$
$\frac{2x+9}{4x}$	$+$		$-$	$\Vert$	$+$

$$x \in \left[-\frac{9}{2}, 0\right)$$

Note 0.4.2, negative case.

$$\begin{aligned} \frac{2x-9}{4x} &\leq -1 \\ \frac{2x-9+4x}{4x} &\leq 0 \\ \frac{6x-9}{4x} &\leq 0 \\ \frac{2x-3}{4x} &\leq 0 \end{aligned}$$

Note 0.4.1, $c \in \left\{0, \frac{3}{2}\right\}$, $x \neq 0$.

		0		$\frac{3}{2}$	
$2x-3$	$-$		$-$		$+$
$4x$	$-$		$+$		$+$
$\frac{2x-3}{4x}$	$+$	$\Vert$	$-$		$+$

$$x \in \left(0, \frac{3}{2}\right]$$

Note 0.4.2, unite intervals.

$$\boxed{x \in \left[-\frac{9}{2}, 0\right) \cup \left(0, \frac{3}{2}\right]}$$

4

Solve the following word problems algebraically.

4.1

Find 3 consecutive even numbers whose sum is 78.

$$=$$

4.2

If the length of each side of a square is increased by 2 units, the resulting square has an area that is 16 units more than the area of the original square. Determine the dimensions of the original square.

$$=$$

4.3

Susan is 9 years older than Ivy. Four years ago, the sum of their ages was 17 years. In how many years will Susan turn 18?

$$=$$

4.4

Ray and Joyce leave a park at the same time in separate directions. Ray drives 6 kilometers home while Joyce drives 7 kilometers home. They arrive at the same time at their respective homes. If Ray drives 3 kilomteres per hour slower than Joyce, what are their average rates?

$$=$$

4.5

To exclusively rent a videoke room, a certain number of people have to pay equal amounts to a total of Php 720.00. If there were 3 persons less, each person would then have to contribute Php 40.00 more. How many people were there originally?

$$=$$

4.6

The numerator of a positive fraction is 5 more than thrice its denominator. If 13 is added to the denominator and 9 is subtracted from the numerator, the resulting fraction is twice the reciprocal of the original number. Find the original fraction.

$$=$$

4.7

Two friends living 39 miles apart leave their homes at the same time on bicycles and travel towards each other. If one person averages 2 miles per hour more than the other and they meet in $\frac{3}{2}$ hours, what is each person's average cycling speed?

$$=$$

4.8

Bus A leaves a terminal at 12:00 noon while Bus B headed for the same destination leaves the same terminal 15 minutes later. After covering 75 kilometers, Bus B catches up to Bus A at 1:15 pm. Determine the speed of each bus.

$$=$$

4.9

One alloy contains 80% gold (by weight) and another contains 55% gold. How many grams of each alloy must be combined to make 40 grams of an alloy that is 70% gold?

$$=$$

4.10

One acid solution is 60% water and another acid solution is 40% water. How many liters of each solution should be mixed to produce 90 liters of acid solution which is 52% water?

$$=$$

4.11

A grocer wishes to mix two types of coffee beans. One costs Php 40.00 per kilogram and the other costs Php 60.00 per kilogram. How much of each type must be mixed to obtatin 50 kilograms of coffee that reasonably costs Php 55.00 per kilogram?

$$=$$

4.12

Wes invested a total of Php 120,000.00 in two accounts. One account earns 7% annual interest while the other earns 5% annual interest. If a year after the investment, Wes earns Php 7,400.00, how much did he put in each account?

$$=$$

4.13

If a rowing team rows with the current, they can cover 16 miles in 2 hours. Against the current, they can cover 10 miles in 4 hours. Find their rate of rowing in calm water and the rate of the current.

$$=$$

4.14

Mary can clean an office in 10 hours. Working together, John and Mary can clean the office in 7 hours. How long would it take for John to clean the office by himself?

$$=$$

4.15

An office has two photocopy machines. Machine A can photocopy 1000 pieces of a single file in 5 hours, while Machine B can photocopy 1000 pieces of the same file in 4 hours. How long will it take the two machines, both working together to photocopy 1600 pieces of a file?

$$=$$

4.16

A man can paint a room by himself in 8 hours while his son can paint the same room in 12 hours. The man works by himself for 3 hours and is then joined by his son. How many hours will it take them to finish painting?

$$=$$

4.17

A woman can do a certain job in 8 hours, while her daughter can do it in 12 hours. After the woman and the daughter worked for an hour, they were joined by the woman's son, and the three completed the job in three hours. How long would it take the son to finish the same job alone?

$$=$$